Chapter 8, Lecture 8

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Normal theory

Throughout, we assume that we sample from a Normal distribution: $X_1, \ldots, X_n$ are iid $N(\mu, \sigma^2)$. We will introduce Student's t_n-distribution and the χ^2_n distribution. Moreover, we will prove that the sample mean and sample variance are independent. Finally, we will derive the exact distributions of the sample variance and the Studentized sample mean. This will allow us to construct exact confidence intervals for μ and σ^2.
An exact $1-\alpha$ c.i. for μ is easily seen to be

$$\left[\overline{X}_n \pm \frac{\sigma}{\sqrt{n}} z(\alpha/2)\right]$$

This is based on the fact that $\overline{X}_n$ has an exact normal $N(\mu, \sigma^2/n)$ distribution. However, often σ^2 is unknown, and then we cannot compute this interval. We can replace it by S^2, the sample variance.
Q: What is the distribution of the Studentized mean

$$\frac{\overline{X}_n - \mu}{S/\sqrt{n}} \ ?$$

A: It was given by Gosset (1908). It is a t_{n-1}-distribution, so all we need to do is to replace the $z(\alpha/2)$ upper points.

χ^2_n distribution

Definition Let $Z_1, \ldots, Z_n$ be iid $\sim N(0,1)$. Then the random variable

$$Y := \sum_{i=1}^{n} Z_i^2$$

has a χ^2_n distribution. We say Y has a chi-square distribution with n degrees of freedom.

Claim: A χ^2_n distribution is the same as a Gamma$(n/2, 1/2)$ distribution.
We will now prove this claim. First we recall the Gamma(α, λ) density

$$f(x) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} \lambda^\alpha \exp(-\lambda x) 1\{x \geq 0\}$$

with

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} \exp(-x)\, \mathrm{d}x.$$

Note that for $\alpha = 1/2$, we have

$$\begin{aligned}
\Gamma(1/2) &= \int_0^\infty x^{-1/2} \exp(-x)\, \mathrm{d}x \\
&= 2\int_0^\infty \exp(-x)\, \mathrm{d}\sqrt{x} \\
&= 2\int_0^\infty \exp(-y^2/2)\, \mathrm{d}\frac{y}{\sqrt{2}} \\
&= 2\sqrt{\pi}\frac{1}{2} = \sqrt{\pi}
\end{aligned}$$

Now, first we consider $n = 1$, so $Y = Z_1^2$. Then the cdf of Y can be written as

$$\begin{aligned}
F_Y(y) &= \mathbb{P}\{Y \leq y\} = \mathbb{P}\{Z_1^2 \leq y\} \\
&= \mathbb{P}\{-\sqrt{y} \leq Z_1 \leq \sqrt{y}\} \\
&= \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) \\
&= 2\Phi(\sqrt{y}) - 1
\end{aligned}$$

and its density as, by taking a derivative,

$$\begin{aligned}
f_Y(y) &= y^{-1/2}\phi(\sqrt{y}) \\
&= y^{-1/2}\frac{1}{\sqrt{2\pi}} \exp(-y/2) \\
&= \frac{1}{\Gamma(1/2)}(1/2)^{1/2} y^{-1/2} \exp(-y/2)
\end{aligned}$$

which we recognize as the Gamma$(1/2, 1/2)$ density. This proves the claim for $n = 1$.
For $n \geq 2$, we reason as follows. Recall that the sum $X + Y$ of two independent random

variables X an Y, with a Gamma (α, λ) and a Gamma (β, λ), respectively, has again a Gamma distribution, with parameters $\alpha+\beta$ and λ. Thus $Z_1^2+Z_2^2$ has a Gamma(1,1/2) distribution, and for general n,

$$Y \sim \Gamma(n/2, 1/2).$$

This proves the claim. ■

We see that Y has mean

$$\mathbb{E}[Y] = \frac{n/2}{1/2} = n$$

and variance

$$\mathrm{Var}(Y) = \frac{n/2}{(1/2)^2} = 2n.$$

Also, we see that if $X \sim \chi_n^2$ and $Y \sim \chi_m^2$, and X and Y are independent, the sum $X + Y \sim \chi_{n+m}^2$.

Student t_n-distribution

Definition Let $Y \sim \chi_n^2$ and $Z \sim N(0,1)$ be independent. Then

$$X := \frac{Z}{\sqrt{Y/n}} \sim t_n.$$

Claim: X has density

$$f(x) = \frac{\Gamma((n+1)/2)}{\Gamma(n/2)}(n\pi)^{-1/2}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2}.$$

Try in R the following:

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curve(dnorm(x,0,1),-4,4)
for (n in 1:100)
curve(dt(x,n),-4,4,col=n+1,add=T)
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We see the following:

- For $n = 1$, we recover

$$f(x) = \pi^{-1}(1 + x^2)^{-1},$$

the Cauchy-distribution.

- For $n \to \infty$, since

$$\left(1 + \frac{x^2}{n}\right)^{-(n+1)/2} \to \exp((x^2 \cdot -\frac{1}{2}) = \exp(-x^2/2)$$

we recover the standard normal $N(0, 1)$ distribution.

- In general, the t_n distribution is symmetric around zero, and has heavier tails than the normal distribution. We have

$$\mathbb{E}[X^k] < \infty \qquad \text{for } k < n$$

and

$$\mathbb{E}[|X|^k] = \infty \qquad \text{for } k \geq n.$$

There exists no moment generating function, and we have $\mathbb{E}[X] = 0$ by symmetry of $f(x)$ around $x = 0$, for $n \geq 2$, and $\mathrm{Var}(X) = n/(n-2)$ for $n \geq 3$. For instance, for $n = 1$,

$$\begin{aligned} \int_0^\infty x \frac{1}{\pi(1+x^2)}\,\mathrm{d}x &\geq \int_1^\infty x \frac{1}{2\pi x^2}\,\mathrm{d}x \\ &= \frac{1}{2\pi}\int_1^\infty \frac{1}{x}\,\mathrm{d}x = +\infty. \end{aligned}$$

Finally, we prove the claim. We use the following trick:
1. Consider the transformation

$$(y, z) \mapsto (x, w) := (z/\sqrt{y/n}, y)$$

2. Compute the joint density of (X, W).
3. Integrate out w *to obtain the marginal density of* X.

Since we have

$$\begin{aligned} y &= w \\ z &= x\sqrt{w/n} \end{aligned}$$

we can compute the determinant of the Jacobian

$$\begin{aligned} det(J(x,w)) &= \frac{\partial y(x,w)}{\partial x}\frac{\partial z(x,w)}{\partial w} - \frac{\partial y(x,w)}{\partial w}\frac{\partial z(x,w)}{\partial x} \\ &= 0 - 1 \cdot \sqrt{w/n} \\ &= -\sqrt{w/n} \end{aligned}$$

and, using the formula at pages 101-102 in your textbook,
2. we compute

$$\begin{aligned} f_{X,W}(x,w) &= f_{Y,Z}(y(x,w), z(x,w))|J(x,w)| \\ &= f_{Y,Z}(w, x\sqrt{w/n})\sqrt{\frac{w}{n}} \\ &= f_Y(w) f_Z(x\sqrt{w/n})\sqrt{\frac{w}{n}} \qquad \textit{by independence} \\ &= \frac{1}{\Gamma(n/2)} w^{\frac{n}{2}-1} \left(\frac{1}{2}\right)^{\frac{n}{2}} \exp(-w/2)\frac{1}{\sqrt{2\pi}} \exp(\frac{1}{2}x^2 w/n)\sqrt{\frac{w}{n}} \\ &= C_n w^{\frac{n+1}{2}-1} \exp(-h(x)w) \end{aligned}$$

with

$$\begin{aligned} C_n &= 1/\left(2^{(n+1)/2}(n\pi)^{1/2}\Gamma(n/2)\right) \\ h(x) &= \frac{1}{2}(1+\frac{x^2}{n}) \end{aligned}$$

3. Finally, compute

$$\begin{aligned} f_X(x) &= \int_0^\infty f_{X,W}(x,w)\,\mathrm{d}w \\ &= \int_0^\infty C_n w^{\frac{n+1}{2}-1}\exp(-h(x)w)\,\mathrm{d}w \\ &= C_n\Gamma((n+1)/2)(h(x))^{-\frac{n+1}{2}} \qquad \textit{recognizing part of a } Gamma((n+1)/2, h(x)) \textit{ density} \\ &= \frac{\Gamma((n+1)/2)}{\Gamma(n/2)}(n\pi)^{-1/2}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2} \end{aligned}$$

as claimed. ■